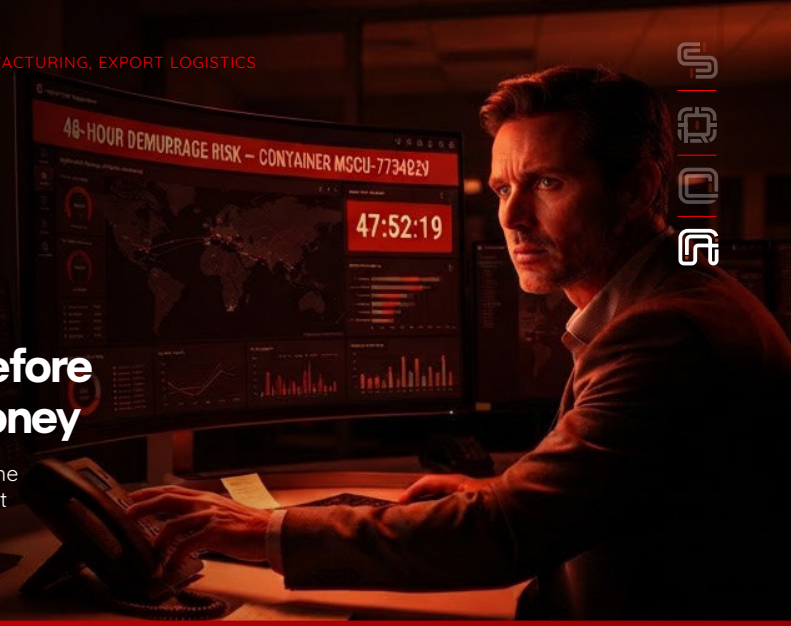
STHIROS
Strategy. Risk. Cyber. AI.



Catching customs and demurrage risk 48 hours before it costs a steel exporter money

A prominent steel manufacturer exporting from India to the United States engaged Sthiros to deploy Keliros, an Export Workflow Management System unifying six global teams and ten document types.

Client	Product Deployed	Sector	Geography
A Prominent Steel Manufacturer Export Logistics	**Keliros, Oriq by Sthiros, Export Workflow Management System**	• **Steel Manufacturing** • **Export Logistics**	**India, United States (Export Trade Lane)**

The Challenge:

When six teams and ten documents move on email and WhatsApp

This manufacturer's export shipments involved over six global teams, India Logistics, Customs House Agents, Freight Forwarders, US Brokers, and Finance, each cycle generating 10 critical document types. An AI-powered OCR engine already extracted data from 800+ fields, but no workflow orchestration layer existed to unify these capabilities into one management tool.

Fragmented Data

Critical documents sat scattered across email, WhatsApp, & disparate CHA/Freight Forwarder portals, with no single source of truth.

Sequence Gaps

No defined SLAs or enforced sequence management existed across six or more internal & external teams.

Manual Validation

Weights, container numbers, and HTS codes were checked manually across 10+ documents, exposed to human error.

Operational Risk

Unvalidated document mismatches, particularly on HTS codes, led directly to US customs delays, additional duties, and shipment holds.

Financial Leakage

A lack of proactive visibility into container status and Last Free Days resulted in avoidable demurrage and detention charges.

Growth Constraint

Manual tracking and ERP journal entries meant scaling shipment volume required a linear increase in headcount.

The Solution:

Keliros: Five Modules, from Sales Invoice to Proof of Delivery

Sthiros deployed Keliros as a five-module Export Workflow Management System, governing the full shipment lifecycle under one digital backbone with blocking gate logic enforcing strict SLAs at every stage.

The 22-week deployment moved the operation from reactive handling to enforced operational discipline, with every document, validation, and alert traceable to source.

The Methodology

1 Sequence Management & Orchestration

A configurable workflow triggered the moment a Sales Invoice is pulled, with blocking gate logic preventing progression until mandatory documents are uploaded and validated.

2 AI-Driven OCR & Cross-Document Validation

Native OCR auto-classifies files and extracts data across 800+ field schemas, validated by 63 rules checking HTS codes and container counts against every prior document.

3 Human-in-the-Loop Document Generation

Six downstream document types generated automatically, with every auto-populated field traceable to sourcae and requiring human approval only as a final check.

4 Real-Time Inventory Lifecycle Tracking

SKU-level inventory tracked via API webhooks across 10 milestones, with alerts tuned to Last Free Days to mitigate detention risk.

5 Accounting Entry Ticketing

Pre-computed accounting entry tickets generated whenever financial documents are validated, mapping debits and credits for ERP posting without added headcount.

What This Means

Catch errors before customs does.

Scale shipments through configuration.

Beat the last free day.

Outcomes

98%+

Field match on auto-generated documents

Eliminating manual re-entry across complex documents like Metal Content Sheets.

63

Cross-validation rules enforced

Checking HTS codes and container counts before submission to US Customs.

95%+

OCR extraction accuracy

The foundation the validation engine runs on, across 800+ data fields.

48Hrs

Advance demurrage alert window

Before the Last Free Day, stopping port penalty leakage before it happens.

Get in touch

your export cycle orchestrated, or just coordinated?

Speak to Sthiros about what Keliros can unlock for your cross-border logistics operation.

Connect with us:
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